JOSEF DOORNINK

Site Reliability Engineer | AI Infrastructure & MLOps

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PROFESSIONAL SUMMARY

A creative problem-solver who creates solutions where none exist, I build the infrastructure that scales research into reality. Few production engineers have published 10 peer-reviewed papers, earned 2 US patents, won a major design award, and shipped FDA-cleared hardware. That discipline now drives 10+ years of SRE and AI infrastructure work (CKS/CKA certified), bridging mathematical rigor and system integrity at scale - currently designing autonomous Agentic SRE pipelines that leverage LLMs for root-cause analysis.

TECHNICAL SKILLS

Core Technologies: Kubernetes (AKS) | Docker | Terraform | Azure | AWS | GCP | Helm | YAML | Python | Go/Golang | Bash | C# | SQL | Kafka | Redis | ElasticSearch

AI/ML Engineering: LLM Model Serving | ML Pipelines | Distributed Training Systems | RLHF Infrastructure | Azure Machine Learning | Vertex AI | Drift Detection | Model Finetuning Systems

Observability & SRE: New Relic | Azure Monitor | Distributed Tracing | CI/CD Pipelines | GitHub Actions | Performance Profiling | Incident Response | Prometheus | Grafana | Azure DevOps

KEY TECHNICAL PROJECTS

K8gentS - What happens when you deploy a non-deterministic reasoning engine in a system that requires guarantees? K8gentS is an autonomous Kubernetes RCA agent built around that question. It routes cluster failures through Gemini-powered analysis, gates remediation behind both a human approval and an OPA Gatekeeper admission policy, and exposes diagnostics via an MCP server published on the official MCP Registry as io.github.JDoornink/k8gents. The hard problems - confidence calibration, LLM vs deterministic routing, evaluating for unanticipated failures - are openly documented in the README.

Listed on the official MCP Registry.

OmniSight-Core - A production-grade video search engine capable of understanding semantic queries (e.g., " Find a red truck at night"). Demonstrates self-healing infrastructure that automatically detects model performance decay and triggers retraining.

agent-lint-cli - ESLint for agents. A published Python CLI tool that validates MCP servers and scans AI agent implementations for security vulnerabilities. Supports configurable security levels, CI/CD integration with threshold-based failure conditions, and multiple output formats including SARIF.

Published on PyPI as agent-lint-cli.

Agentic SRE Pipeline & Portfolio - The source code driving this exact platform. A Next.js (React) infrastructure executing a Python/RAG Agent pipeline that strictly parses unstructured Job Descriptions and outputs statically generated, targeted frontend bundles dynamically.

CERTS/COURSES

- **CNCF Certified Kubernetes Security Specialist (CKS)** | CNCF | March 2024
- **CNCF Certified Kubernetes Administrator (CKA)** | CNCF | June 2021
- **Machine Learning Specialization** | Stanford / Coursera | September 2025
- **HashiCorp Certified Terraform Associate** | HashiCorp | July 2022
- **Microsoft Certified Azure Developer Associate** | Microsoft | August 2019
- **Production Machine Learning Systems** | Google Cloud / Coursera | April 2026

PROFESSIONAL EXPERIENCE - STARTUP

Lead MLOps Engineer | REASON BENEFIT AI CORPORATION | October 2025 - Present

- Architect and maintain large-scale Azure Kubernetes Service (AKS) production environment for ML model training and serving, supporting distributed model inference at scale.
- Integrate with Azure offerings for resource utilization across distributed training systems.
- Collaborate with research teams to translate desired microservice architectures into cloud-based systems with focus on reliability and scalability.
- Drive distributed training pipeline creation by introducing automated pipeline scripts and validations to reduce training cycle time.

PROFESSIONAL EXPERIENCE

Lead Site Reliability Engineer (SRE) I -> II -> III | Trimble/Viewpoint | January 2019 - Present

- Built and maintained Human Resources Information management system, responsible for over $8.3M+ in ARR and 43% YoY growth rate.
- Architected AKS production environments handling 14K+ requests/day across 30+ microservices at 99.9% uptime SLA.
- Built automation tooling in Python and Go eliminating 80+ hours/month of toil and accelerating deployment velocity 3x.
- Developed YAML-based orchestration pipelines, reducing manual overhead by 70%.
- Reduced P99 latency 45% and improved throughput 60% through systematic profiling of distributed systems.
- Developed Go CLI tooling (Cobra) adopted by 50+ engineers for streamlined infrastructure workflows in the organization.
- Implemented New Relic observability stack with distributed tracing, cutting MTTR by 50%.
- Led Kubernetes capacity planning and auto-scaling strategies supporting 200% traffic growth.
- Built CI/CD pipelines (GitHub Actions, Azure DevOps) with automated testing and rollback mechanisms.
- Managed 100+ cloud resources via Terraform IaC; implemented CKS security controls for SOC2 compliance.
- Integrated services with event-driven architecture using Kafka and Redis, improving system decoupling and scalability.
- Mentored junior engineers and led knowledge-sharing sessions on Kubernetes best practices and SRE principles.
- Maintained pipelines across multiple teams and Cloud platforms (Azure, AWS) for consistency across products in the organization.

Software Developer | Viewpoint | March 2018 - January 2019

- Developed cloud-based SaaS applications using .NET and Angular, migrating on-premise software solutions to Azure cloud platform.
- Built RESTful APIs for multi-tenant applications serving thousands of users with focus on performance and scalability.

Software Developer I | Onfulfillment | March 2014 - March 2018

- Engineered multi-tenant e-commerce platform using Microsoft Stack (.NET, C#, SQL Server) integrated with third-party SaaS APIs.
- Led 'uplift' initiative migrating legacy codebase to modern greenfield platform, improving response times by 40% measured through New Relic APM.

Biomechanical Research Engineer II | Legacy Biomechanics Research Lab | 2007 - 2013

- Lead Test and Development Engineer for NIH-funded multimillion-dollar research project focused on bone fixation solutions.
- Managed successful implant creation, delivery, and test methodology producing multiple US FDA-approved implants.

EDUCATION

Master of Science, Biomedical Engineering - University of California, Davis | 2006

Bachelor of Science, Mechanical Engineering - California State University, Chico | 2003

PATENTS & AWARDS

Bone screw with multiple thread profiles for far cortical locking and flexible engagement to a bone (US10918430B2)

- Continuation patent for the far cortical locking bone screw. A bone screw with dual thread profiles enabling flexible engagement to the far cortex only, allowing dynamic fixation that promotes superior fracture healing.

Bone screw with multiple thread profiles for far cortical locking and flexible engagement to a bone (US9314286B2)

- A bone screw including a head, a shaft, and a tip. The shaft includes a first threaded section, a second threaded section, and an unthreaded section positioned between the first and second threaded sections. Designed to enable dynamic fixation.

Bone screw with multiple thread profiles... (US8740955B2)

- Original patent formulation for the orthopedic implant enabling far cortical locking via a flexible-engagement thread profile.

Scientific Exhibit Award of Excellence

- Awarded at the American Academy of Orthopaedic Surgeons (AAOS) 2010 Annual Meeting for the landmark exhibit: 'Effects of Construct Stiffness on Healing of Fractures Stabilized With Locking Plates' (Bottlang M., Doornink J., Fitzpatrick DC, Marsh JL, Augat P., von Rechenberg B., Lesser M., Madey SM).

PUBLICATIONS

- M Bottlang, **J Doornink**, DC Fitzpatrick, SM Madey. *Far cortical locking can reduce stiffness of locked plating constructs while retaining construct strength.* The Journal of Bone and Joint Surgery (JBJS), 2009
- M Bottlang, **J Doornink**, GD Byrd, DC Fitzpatrick, SM Madey. *A nonlocking end screw can decrease fracture risk caused by locked plating in the osteoporotic diaphysis.* The Journal of Bone and Joint Surgery (JBJS), 2009
- DC Fitzpatrick, **J Doornink**, SM Madey, M Bottlang. *Relative stability of conventional and locked plating fixation in a model of the osteoporotic femoral diaphysis.* Clinical Biomechanics, 2009
- M Bottlang, M Lesser, J Koerber, **J Doornink**, B von Rechenberg, P Augat, DC Fitzpatrick, SM Madey, JL Marsh. *Far cortical locking can improve healing of fractures stabilized with locking plates.* The Journal of Bone and Joint Surgery (JBJS), 2010
- M Bottlang, **J Doornink**, TJ Lujan, DC Fitzpatrick, JL Marsh, P Augat, B von Rechenberg, M Lesser, SM Madey. *Effects of construct stiffness on healing of fractures stabilized with locking plates.* The Journal of Bone and Joint Surgery (JBJS), 2010
- **J Doornink**, DC Fitzpatrick, S Boldhaus, SM Madey, M Bottlang. *Effects of hybrid plating with locked and nonlocked screws on the strength of locked plating constructs in the osteoporotic diaphysis.* Journal of Trauma and Acute Care Surgery, 2010
- **J Doornink**, DC Fitzpatrick, SM Madey, M Bottlang. *Far cortical locking enables flexible fixation with periarticular locking plates.* Journal of Orthopaedic Trauma, 2011
- PJ Denard, **J Doornink**, D Phelan, SM Madey, DC Fitzpatrick, M Bottlang. *Biplanar fixation of a locking plate in the diaphysis improves construct strength.* Clinical Biomechanics, 2011
- PA Wackym, JA Ratigan, JD Birck, SH Johnson, **J Doornink**, M Bottlang, SK Gardiner, FO Black. *Rapid cVEMP and oVEMP responses elicited by a novel head striker and recording device.* Otology & Neurotology, 2012
- M Bottlang, DC Fitzpatrick, D Sheerin, E Kubiak, R Gellman, C Vande Zandschulp, **J Doornink**, K Earley, SM Madey. *Dynamic fixation of distal femur fractures using far cortical locking screws: a prospective observational study.* Journal of Orthopaedic Trauma, 2014